

Karen Shieh

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EDUCATION

University of California, Berkeley

Berkeley, CA

M.S. in Electrical Engineering and Computer Sciences

Aug. 2026 – Expected May 2027

- Research interests: computer security, systems, and security log analysis

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science, B.A. in Data Science | R&C Scholar | 4.00 GPA

Aug. 2022 – May 2026

- Related coursework: Data Structures, Efficient Algorithms and Intractable Problems, Machine Structures, Introduction to Machine Learning, Optimization Models in Engineering, Designing/Visualizing/Understanding Deep Neural Networks, Operating Systems and System Programming, Natural Language Processing, Security in Computer Systems, Deep Reinforcement Learning, Decision Making, and Control

EXPERIENCE

Undergraduate Student Researcher

Jan. 2025 – Present

University of California, Berkeley (REDS)

Berkeley, CA

- Fuzzing: Developed an LLM-driven pipeline that automatically scrapes seed files from the web, including GitHub repositories, to generate off-the-shelf fuzzing corpora and lower the entry barrier to fuzz testing.
- Log Detection: Built an iterative LLM-assisted workflow for generating, testing, and refining security log detection rules from example malicious and benign system logs.
- Vulnerability Detection: Built browser-based testing tools to detect security-relevant behavior changes in web applications introduced by code modifications.

Software Engineering Intern

May 2025 – August 2025

ClickTime

San Francisco, CA

- Developed C#/.NET backend APIs, created SQL tables with triggers, and built React frontend components for Time Entry Activity, a heatmap dashboard to visualize employee time entry behavior
- Designed a reusable and scalable full-stack architecture to ensure maintainability and support future extensions
- Built user-centric features that streamlined workflows and integrated with existing systems
- Leveraged GitHub Copilot to accelerate development efficiency and achieve stretch goals beyond the original project scope

Web Development Assistant

May 2024 – May 2025

Safe Transportation Research and Education Center

Berkeley, CA

- Built a dynamic SvelteKit website for Southern California cities to collect and manage data on bicycle, pedestrian, and wheelchair traffic for infrastructure projects
- Designed and maintained relationships between tables in the database through Prisma and SQL commands

PROJECTS

Estuary

Aug. 2024 – Present

- Develop full-stack features for an open-source multimodal framework for building low-latency, real-time voice AI agents ([GitHub](#))
- Built and shipped client SDKs and demo apps across 4 platforms: Unreal Engine (5.6–5.8), Lens Studio (Snap Spectacles), Unity, and TypeScript/web, including docs, example projects, and release validation
- Developed RAG knowledge-bank features (PDF ingestion, pgvector retrieval) with an LLM-as-judge benchmark harness to evaluate retrieval quality
- Engineered LiveKit/WebRTC session lifecycle management, secure character sharing, Stripe billing, and 5+ LLM/TTS provider integrations (Gemini, Inworld, MiniMax, ElevenLabs)

TECHNICAL SKILLS

Languages: Java, Python, C/C#, SQL, Scheme, JavaScript, TypeScript, HTML/CSS

Frameworks: React, Svelte/kit, Unity Game Engine, Jupyter Notebook, HoloLens 2, Qualtrics, Prisma, .NET

Developer Tools: Git, Docker, VS Code / Copilot, Visual Studio, PyCharm, Cursor, Google Colab, SSMS, Postman

Libraries: pandas, NumPy, Matplotlib, seaborn, PyTorch

Certifications: Cloud Computer Basics, Introduction to Programming with MATLAB, New Models of Business in Society, CAD Mechanical Design I, Computer Essential Skills & Programming